

1. Hello, Everyone. This is Dr. Y. Zhang. In this session, we'll learn about cardiac action potentials. Here's my email in case you need it.
2. Here are the session objectives. These are the contents you are expected to learn.
3. This is the heart. As you know, the heart has four chambers. The coordinated contraction and relaxation of the four chambers are made possible by these specialized cardiac pacemaking and conduction system. The normal heartbeat is initiated inside the sinoatrial node (or sinus node) and transmitted to the rest of the heart in the forms of cardiac action potentials. Cardiac action potentials are the basis for all cardiac electrophysiology, including the EKG you will learn. It's also important in understanding the mechanisms of arrhythmia formation and antiarrhythmic medications, and all the way to clinical cardiac electrophysiology.
4. This slide shows how the cardiac action potential is recorded. You have to insert a microelectrode into a myocyte, and the electrode is connected to the positive side of the voltage meter. Another electrode is placed outside of the cell, and it goes to the negative side of the meter. So, the action potential is a recording of transmembrane voltage differences inside the myocyte versus outside during a cardiac cycle. In other words, it's looking from inside the myocyte. In resting state, the intracellular potential is negative, about -90 millivolts in atrial and ventricular myocytes. Pacemaker cells in the sinus node and the AV node have lesser negative values, and we'll discuss those later.
5. Here, I'm showing you an isolated ventricular myocyte, how it looks like under the microscope.
6. Shown here are the action potentials recorded from different parts of the heart. From top going down are action potentials from the sinus node, the atria, the AV node, His bundle, bundle branches, Purkinje fibers, and the ventricular myocytes at the bottom. Obviously, they have different shapes and duration.
7. Here, I'm listing all cardiomyocytes. On the top are the pacemaking and conduction system, the atrial and ventricular myocytes are at the bottom. Cardiac action potentials are divided into two types, the fast response action potential, and the slow response action potential. As you can see, only the sinus node and the AV nodal fibers generate slow response action potential. All other myocytes generate fast response action potential. Please remember this.
8. Here are the major differences of the two types of action potential. Fast response versus slow response action potential. First, phase zero depolarization is

caused by opening of the sodium channels in fast response action potential, versus the L-type calcium channels in slow response action potential. Second, the rising speed, one is very fast versus the other is slow. In fact, this is how they were named. In addition, there are other differences we'll learn later.

9. Before going into the details of the action potentials, we'll first review some important elements. First, there are three ions, potassium, sodium, and calcium, you should remember. These are the major players in generating cardiac action potentials. Secondly, you have to know that these ions are not free to cross cell membrane. They have to go through their specific channels or gates in order to get in or get out of the cell. In other words, if the channels are closed, they cannot cross the cell membrane. Finally, there are three pumps you must remember. The sodium potassium ATPase, the sodium calcium exchanger, and the calcium ATPase. The name ATPase indicates they need energy, ATP, in order to operate. The first two pumps are located on cell membrane, and the calcium pump is mainly located on the membrane of the sarcoplasmic reticulum (SR) inside the myocyte. These pumps are important in maintaining electrolyte homeostasis.

10. First, let's look at the potassium ion. Assuming this is the myocyte. First thing you should know is that the potassium concentration inside the cell is very high. It is much higher than the outside. The potassium concentration in extracellular fluid is equivalent to the concentration in blood. The normal blood potassium concentration is about 4 mM, in the range somewhere from 3.5 to 5 mM. While inside is much higher, it's 140-150 mM. Due to these concentration differences, when potassium channels open, the potassium ion will flow out along the chemical gradient. Here is an important question for you. Assuming potassium channels stay open. Will potassium ions keep going out until the concentration inside the cell equals to the outside. I will give you a few seconds to think.

The correct answer is no. Here is the reason. The potassium ion carries positive charges. As more potassium ions leave the cell, they also bring out more positive charges. Inside the cell, a negative potential is being built up. The negative potential inside the myocyte will try to retain the positively charged potassium ion. Thus, there are two opposing forces here. The chemical gradient will try to push out the potassium while the negatively built-up potential inside the cell will try to retain the potassium. When these two opposing forces become equal, then there will be no net movement. This potential is called equilibrium potential, and it can be estimated using this Nernst equation. Here “z” is the

number of charges. For potassium, it's one. "I" stands for inside concentration. "O" stands for outside concentration. For the given potassium concentration here, the equilibrium potential for potassium is -96 mV. Of course, the number could differ if different potassium concentration is assumed.

11. Next, we'll look at sodium. Here, you have to know that extra cellular sodium concentration is much higher than inside. When the sodium channels open, sodium will rush inside due to the chemical gradient. However, as the positively charged sodium accumulates inside the cell. They build up positive potential. And eventually, the positive potential will prevent sodium from getting into the cell. Similarly, we can use the Nernst equation to calculate the equilibrium potential for sodium. The value here is positive 52 millivolts. In other words, when the inside potential reaches positive 52mV, it will prevent sodium from coming in.

12. Finally, we'll look at calcium. Here you have to remember that extracellular calcium concentration is much higher than inside. Intracellular calcium concentration is very low in cytoplasm. Ten thousand times lower than outside. When calcium channels open. Calcium will move inside due to the chemical gradient. It also brings positive charges inside. Again, you can use the Nernst equation to calculate the equilibrium potential. Here you have to remember, z is 2 for calcium, because it carries two positive charges. For calcium, its equilibrium potential is 134mV.

13. Here is the member potential equation. Practically, we consider only these three major ions, potassium, sodium, and calcium, because other cations and anions play only minor roles. I'm not going to discuss this equation in detail. As shown in the bottom, for any moment, membrane potential equals relative conductance of potassium times its equilibrium potential, add relative conductance of the sodium times its equilibrium potential, then add relative conductance of the calcium times its equilibrium potential. Just give you an example, during resting state, both sodium and calcium channels are closed. There's almost no conductance for sodium and calcium. There are zero. Only potassium channels are open. Theoretically, the membrane potential equals potassium equilibrium potential, which is -96 millivolts. In real life, it's very close, about -90 millivolts, due to variations of K concentration and background leakages.

14. During cardiac cycle, sodium and calcium ions will move inside the cell, and the cell will lose potassium ions. These pumps will work hard to restore electrolyte homeostasis. First, is the sodium potassium ATPase. It transports sodium out and takes in potassium in a ratio of three to two. In other words, it pumps out more positive charges. So, when this pump works, it will bring down the membrane potential (the membrane potential becomes more negative). Another pump is the sodium calcium exchanger. Normally, it pumps out calcium in exchange letting in the sodium in a ratio of three to one. When this pump works, it tends to increase the membrane potential. This is known as the sodium calcium exchanger current. Please remember this. This current will depolarize the cell and could lead to arrhythmia formation. We will discuss this in future lectures. The sodium calcium exchanger normally works to pump out the calcium, as we just discussed. This is known as the forward mode. This pump could work in a reversed mode, meaning it pumps out sodium and lets in the calcium. On which mode it will work depends on the intracellular calcium and sodium concentration. In condition of increased intracellular sodium concentration, such as in the condition of digitalis treatment, the exchanger will try to pump out sodium and let in calcium. Thus, it will increase calcium concentration inside the cell. This is the mechanism the medication digitalis increases cardiac contractility, and you will learn this in pharmacology. The third pump is the calcium ATPase. The calcium ATPase are mainly located on the sarcoplasmic reticulum (SR). Their role is to transport calcium back from cytosol into SR.

15. With previous knowledge, now we are ready to discuss fast response action potentials. You should know which myocytes generate fast response action potential. Fast response action potential has five phases as shown here. Phase zero, phase one, phase two, three, and four. This drawing represents a ventricular action potential. Phase zero, as shown here, is the rising phase, known as depolarization phase. Phases one, two, and three are repolarization phases, and phase four is the resting membrane potential. These color lines indicate different ion currents during each phase. Here is the rule for naming the current. Current is abbreviated using the letter I, either the lower case or upper case can be used. Followed by the type of the ion, potassium, sodium or calcium ion. Then followed by the subtypes of this particular ion current. For example, as shown here, I_{Kto} is the potassium current, “to” stands for transient outward potassium current. We will discuss in detail in the next few slides.

16. Here we'll start with phase zero depolarization. In this phase, the membrane potential rises from resting potential, about -90 millivolts to around positive 20 to 30 millivolts. Fast response cells (like atrial or ventricular myocytes), typically don't generate action potential themselves. Normally, it's being driven by action potential coming from the sinus node. When action potential from the sinus node arrives at the cell, it increases the potential. When the membrane potential reaches the threshold, as shown here, it triggers opening of the sodium channels and generate phase 0 depolarization. Phase 0 depolarization is caused by opening of the fast sodium channel, and subsequently the influx of the sodium current.

17. Here, it's important to understand the voltage gated fast sodium channel status. As shown here, the sodium channel has two gates. The outside activation gate or fast gate, and the inside is the inactivation gate or slow gate. The sodium channel is a voltage sensor. In resting state, the outside activation gate is closed. Inside inactivation gate stays open. When the membrane potential rises to reach its threshold, about -55 millivolts, it triggers opening of the activation gate, which opens very fast. At the same time, it also triggers closing of the inactivation gate or slow gate. By the name you know, it closes slowly. Thus, for a very short, brief period, in a fraction of a millisecond, both gates are open, and the sodium ions rush in, causing depolarization. This is the activation state. When the slow gate closes. It ends phase zero depolarization. This is the inactivated state. In inactivated state, the sodium channel cannot reopen. Note that the sodium channel will stay in inactivated state until membrane potential drops below the threshold level at the end of phase 3, Then the fast gate will close, and the slow gate will slowly open, returning to the resting state and ready to reopen.

18. After depolarization, the next is phase 1, early repolarization. In this phase, the major event is the opening of the transient outward potassium channel, and the potassium ions leave the cell. Phase one repolarization is a modest drop of potential from its peak value around positive 20 to 30 mV to the potential level at the start of the plateau phase, which is around zero millivolt. Since the transient outward potassium channel opens for only a brief time, closure of the potassium channel terminates phase one repolarization.

19. The next is phase 2 or plateau phase. Plateau phase of the repolarization is unique to cardiac fast response cells. In phase two, the potential falls very slowly, almost flat. This is due to the result of balanced inward and outward ion

movements. The major inward ions are calcium in phase 2. Depolarization causes voltage gated L-type calcium channel to open, but they open slowly and last long. The inward calcium current occurs during whole phase two. The outward ion current is potassium. The major potassium channel responsible for outward movement in plateau phase is the delayed rectifier potassium channel. The “delayed” in name indicates that the channel opens slowly. These channels reach their maximum opening at the end of phase two or start of phase three. In fact, the delayed rectifier potassium channel has three components as shown here. I_{Kr} indicating rapid, I_{Ks} , slow, I_{Kur} , ultra rapid, depending on their activation mode. It's good to know these names, you will see these names in literature. The plateau phase is not completely flat; the potential slowly and gradually decreases throughout the plateau phase. So, the outward current is somewhat greater than the inward current.

20. Phase three is the rapid repolarization that follows the plateau phase. When the L-type calcium channels start to close and the delayed rectifier potassium channels reach their maximum opening, phase three is initiated. As member potential drops below -30 millivolts, a second potassium channel called inward rectifying potassium channel opens, which helps to decrease the member potential to its resting level. Most potassium channels close as the member potential approaches its resting value. However, the inward rectifying potassium channel, known as I_{K1} remains open at resting membrane potential. That's why, as we mentioned before, the resting membrane potential is very close to the potassium equilibrium potential.

21. Phase four is the resting membrane potential. In this phase, sodium, calcium channels are closed. Only I_{K1} channel remains open, as we just mentioned. This ensures membrane potential stays near its equilibrium potential for potassium. During phase four, as mentioned before, these three pumps will be working to move extra sodium and calcium out and bring back the potassium into the cell. So, the cell can maintain its normal electrolyte homeostasis.

22. This drawing summarizes the different currents during different phases of the fast response AP. The last 2 currents we have not learned yet and will be discussed later.

23. Here is the minimum requirement for fast response action potential. The five phases and the associated currents.

24. Remember, the heart is very sensitive to blood potassium concentration. Here we'll look at how blood potassium concentration could affect action potentials. First, we'll look at hyperkalemia. Hyperkalemia has 2 major effects: First one is that it increases membrane potential. According to the Nernst equation, when the blood potassium concentration increases from normal four millimolar to seven millimolar, the equilibrium potential, or the resting membrane potential will increase. This can be calculated using the equation. Depending on the level of the rising member potential, it can have the following effect. First, when the member potential rises, it can shorten the distance from the resting state to the threshold, meaning the cell will be easy to be excited (increases its excitability). Second, the increased membrane potential will affect sodium channel recovery, and you should know why. Because sodium channel recovery requires the membrane potential to drop below the threshold. If the member potential stays very close to its threshold, the recovery will be slow. Finally, if the membrane potential cannot drop below the threshold, in other words, it stays above the threshold, then the sodium channel cannot recover. Slow recovery of the sodium channel will affect phase zero depolarization, thus slowing the ventricular depolarization and increasing the interval of the QRS complex on the EKG. If sodium channel cannot recover, then the heart cannot be excited again and will stay in cardiac arrest.

The second change, you have to remember, is that high potassium will open more potassium channels. This will speed up repolarization. The faster repolarization will produce peaked T wave on EKG. You will learn this in EKG lecture.

Here are the potential EKG changes caused by hyperkalemia: peaked T wave, widened QRS complex, and eventually ventricular fibrillation or cardiac wrist.

25. As we just discussed, if the resting membrane potential can't drop below the threshold level for the Na channel, the Na channel will stay in inactivated state and can't re-open. Since the threshold is around -55mV, do you know what blood K concentration is needed to move the membrane potential to this level? You can calculate using the Nernst equation or just ask AI to do this. I asked Copilot to calculate this, and the K concentration is around 17-18mM. Note that the values could slightly change depending on the assumption of intracellular K level (140-150mM).

Since hyperkalemia can lead to cardiac arrest, this is utilized clinically to arrest the heart during open heart surgery or heart transplantation by applying the

cardioplegia solution. On the other hand, you need to know that to avoid rapid increase in blood K concentration, K cannot be given as a bolus IV injection in patients. Please remember this.

26. Now, we'll look at the other extreme, hypokalemia. If the potassium concentration is below normal, by the Nernst equation, the membrane potential will become more negative, as shown here. Theoretically, this could decrease excitability. However, the major risk in hypokalemia is not due to more negative membrane potential, it is due to decreased potassium conductance. In hypokalemia, less potassium channels will be open. This will slow down the repolarization process and prolong action potential duration. This will lead to flattened T waves and long QT intervals on EKG (you'll learn this in EKG lecture). The long QT interval can increase the risk of developing ventricular tachyarrhythmias. Now you should understand why clinically potassium concentration is closely monitored and why hyper or hypokalemia can be diagnosed with EKG.

27. This slide shows the effect of different K concentrations on cardiac AP duration and EKG. The normal potassium level is shown in the middle. When you have low potassium concentration, it opens less K channels, slows down repolarization, thus, increases the action potential duration compared with normal. On EKG, you'll see flattened T wave and the QT interval will be longer. On the other hand, if the potassium concentration is higher than normal, it opens more K channels, the repolarization will be faster, and the action potential duration will be shorter. On EKG you will see peaked T wave, as shown here. Note that ventricular myocytes from epicardial, endocardial and mid-layer have different action potential duration. This will be discussed in future lectures.

28. Another important concept is so-called excitation-contraction coupling. This is the process from action potential propagation leading to cell contraction. In cardiac fibers, there is a system called T-tubules, which extend cell membrane deep into the cell. The action potential, which propagates along cell membrane, can go deep into the cell along the T-tubules. There are voltage gated L-type calcium channels, also known as the dihydropyridine receptors on T-tubules. Opening of the L-type calcium channel permits the calcium ions to go inside. The influx of the calcium from the t-tubules will bind to the calcium release channel on sarcoplasmic reticulum (SR). This channel is also known as ryanodine receptor. The binding of calcium to calcium release channel opens the channel, permitting calcium from SR to be released into the cytoplasm. You should

know that the calcium released from SR is the major source of calcium increase in systole. The spike of calcium concentration in cytosol triggers myocyte contraction. This process is known as the calcium induced calcium release or CICR. Calcium ions are removed from cytosol in diastole by two pumps, causing relaxation. First is the calcium ATPase. It's also known as Sarco(endo)plasmic reticulum Ca-ATPase (SERCA), which pumps calcium back into SR. The other mechanism is the sodium calcium exchanger, which is located on the cell membrane and moves calcium out of the cell.

29. Now, we'll look at slow response action potential. Slow response action potential is simpler, divided into three phases. Phase zero, phase three, and phase four. This drawing represents an action potential from a sinus node fiber. The depolarization phase, phase zero, repolarization phase, phase three and the resting membrane potential, phase four. The color lines represent different currents during different phases. Note that the slow response action potential has lesser negative member potential, around -60 millivolts. And the peak action potential reaches about zero millivolt instead of plus 20 to 30 millivolts in fast response action potential. In addition, phase four in slow response fibers is not flat. It shows spontaneous depolarization.

30. We'll start with phase zero depolarization. Remember, phase zero depolarization in slow response fibers is caused by opening of the L-type calcium channel, instead of the fast sodium channel, seen in fast response fibers. In fact, the slow response fibers don't have voltage gated fast sodium channel. The calcium influx through the L-type calcium channel is relatively slow compared with fast sodium current, see in fast response fibers. As mentioned, the names fast and slow were used to describe the differences of phase zero depolarization in the two types of fibers.

31. Phase three repolarization is caused by opening of the potassium channels, the delayed rectifier potassium channels, and the inward rectifying potassium channel as discussed in fast response fibers.

32. Phase four is resting member potential. Slow response fibers don't have true resting member potential. After repolarization reaches its maximum diastolic potential. The slow fibers begin to spontaneously depolarize, instead of flat, and eventually generate their own action potential. The spontaneous depolarization is mainly initiated by pacemaker channel. It has another name, hyperpolarization-activated cyclic nucleotide-gated channel, or HCN channel, another name is “funny channel”. Its current is called pacemaker current or funny current, If. In

addition, deactivation of I_{K1} channels and activation of T-type Ca channels also contribute to phase 4 spontaneous depolarization.

33. Here are the minimum requirements of slow response action potential. The pacemaker channel and pacemaker current for spontaneous depolarization in phase 4, the L-type Ca current for phase 0 depolarization and K current for phase 3 repolarization.

34. So far, we have learned the fast and slow response cardiac action potentials. This slide shows the differences of action potential duration between nerve/skeletal muscle, and the cardiac action potential. Nerve action potential is very brief (lasts only a millisecond), compared to cardiac action potential, which lasts typically 300 to 400 milliseconds. What's the importance behind the prolonged cardiac action potential? The action potential duration is related to its refractory period. The longer the action potential duration, the longer the effective refractory period (ERP).

35. The longer cardiac action potential duration ensures longer effective refractory period as shown here. As you know, there is an effective or absolute refractory period in action potential, during which a second stimulus, regardless of its strength, cannot evoke another action potential. As we have discussed already, the effective refractory period is determined by the status of fast sodium channel in fast response fibers. This period is from phase zero depolarization to phase three of repolarization. Once the membrane potential drops below its threshold, the sodium channels start to recover, marking the start of relative refractory period. The relative refractory period ends at the beginning of phase four. In relative refractory period, a stronger than normal stimulus can evoke a second, typically smaller action potential because sodium channels are not fully recovered yet in relative refractory period. This panel shows the sodium channel status you have learned.

36. The cause of effective refractory period in slow response fibers is the time required for calcium channels to return to their resting conformation. The relative refractory period in slow response fibers typically extends well into phase four, rather than ending at the beginning of phase four, like in fast response fibers.

37. Effective refractory period is an important concept in cardiac arrhythmia formation and termination. You have to remember this. For example, a type of antiarrhythmic medication, known as the potassium channel blockers, can increase AP duration and ERP.

38. What's the physiological importance of long cardiac effective refractory period? Let's look at this slide. This is the skeletal muscle action potential, and the muscle contraction. Skeletal muscle action potential lasts only a few milliseconds. The muscle contraction occurs well after the effective refractory period. Thus, during muscle contraction, many more action potentials can be generated, and this is the reason tetanus contraction can occur in skeletal muscle.

39. Here is the case in cardiac muscle. The long plateau phase and the long effective refractory period prevent the occurrence of a second action potential during cardiac contraction and a major part of relaxation. Thus, preventing the summation seen in skeletal muscle. This is the reason that cardiac muscle won't develop tetanus contraction.

40. Here is the summary slide. These are the contents you have learned and will be tested.

41. Here I'm giving you some example questions you can check on yourself.

42. Finally, here is the link you can use to submit your feedback. Any comments are welcome and thank you for your attention.